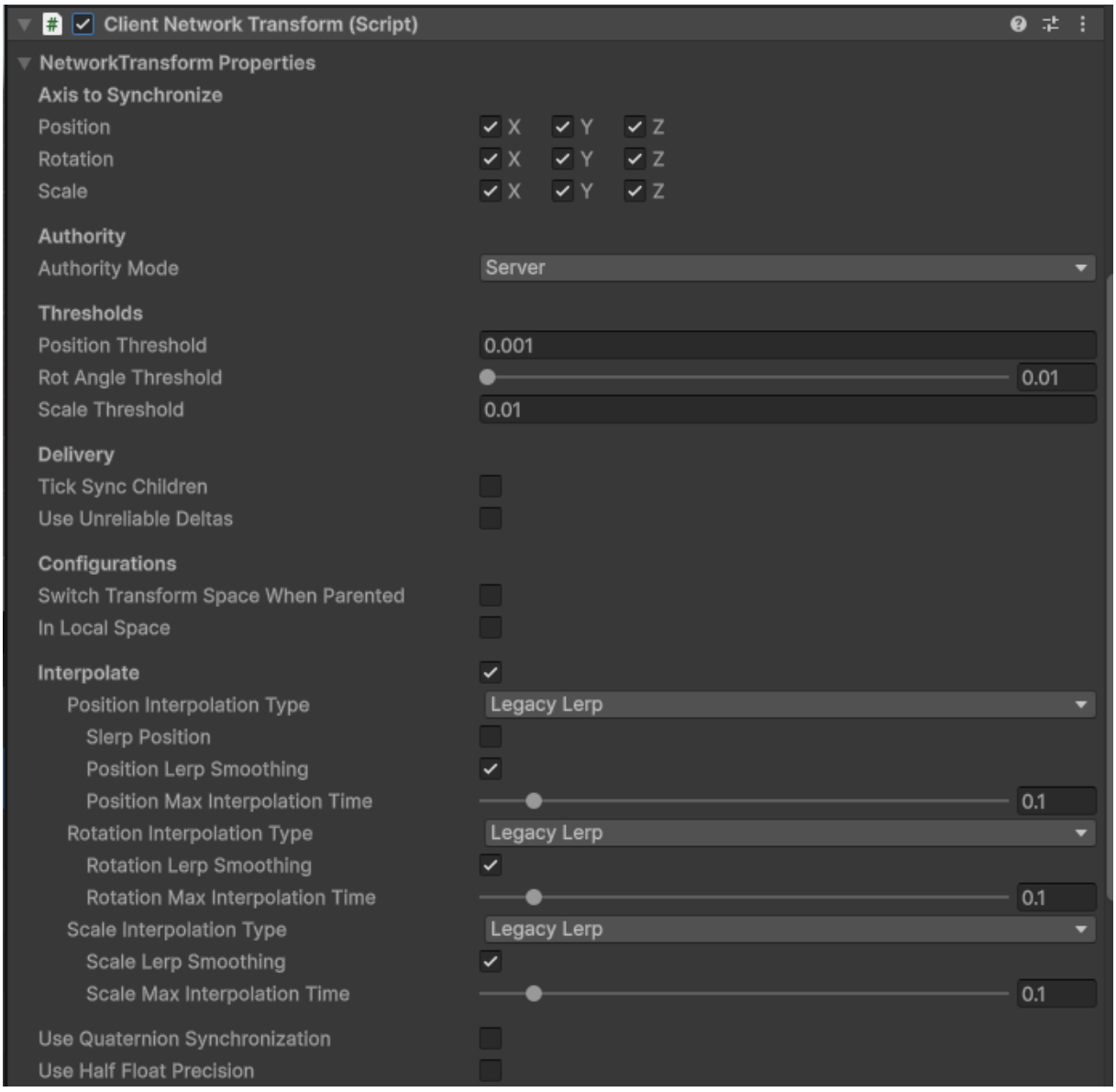
Synchronization guide

This is a step-by-step guide to synchronization using Unity Netcode. It covers NetworkTransform, NetworkAnimator, and NetworkVariables.

NetworkTransform

1. Navigate to the Project window and go to Assets → Objects. Find and select HoverVehicleRoot.
2. Check Client Network Transform box. If there is no client network transform, add a new component called NetworkTransform.
3. Configure the settings as shown in the image.



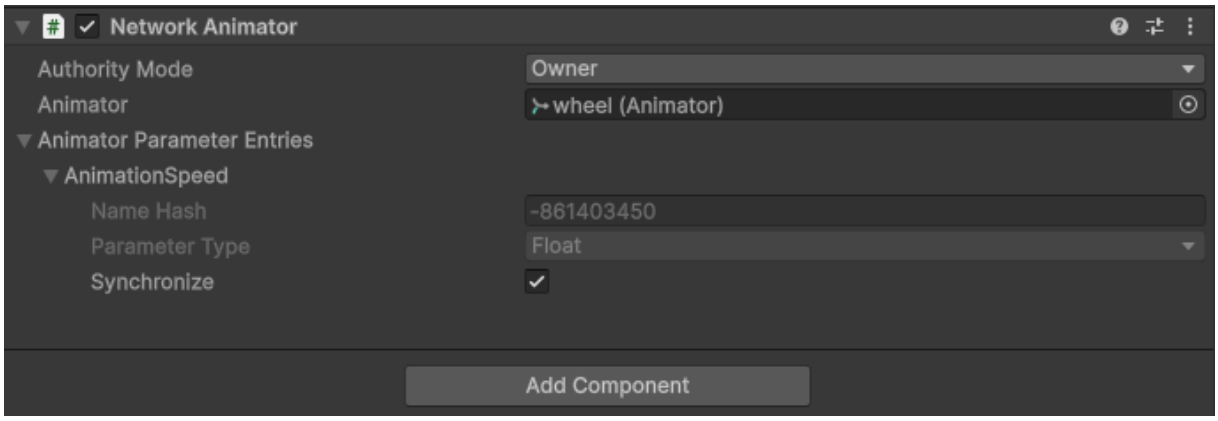
The movement synchronization is now set up, and you are ready to test it.

4. Go to Window → Multiplayer → Multiplayer Play Mode.
5. Add a second player (Player 2).

6. Run de game in de main editor
7. Test the movement synchronization between both players.

NetworkAnimator

1. Navigate to the Project window and go to Assets → Objects. Find and select HoverVehicleRoot.
2. Use the supplied NetworkAnimator; do not add a second one.
3. Configure the settings as shown in the image.



4. Look in the Inspector and open the script HoverVehicleController
5. Add unity.netcode.components like this:

```
using TankDemo.Networking;
using Unity.Netcode;
using UnityEngine;
using UnityEngine.InputSystem;
using Unity.Netcode.Components;
```

6. Change Animator to NetworkAnimator

```
25
26      [Header("Animation")]
        2 references
27      [SerializeField] Animator animator;
28
```

To

```
25
26      [Header("Animation")]
        3 references
27      [SerializeField] NetworkAnimator networkAnimator;
```

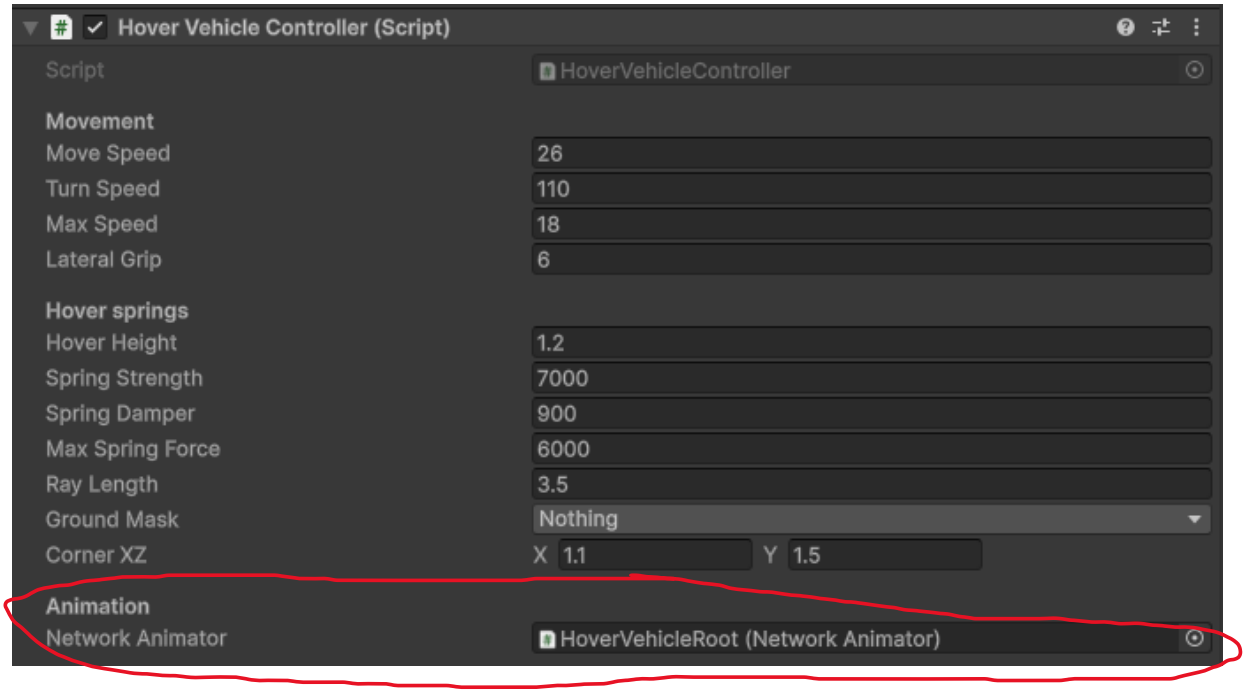
7. Change this

```
132         if (animator != null)
133         {
134             animator.SetFloat("AnimationSpeed", moveInput);
135         }
136
137         // if (networkAnimator != null && networkAnimator.Animator != null)
138         // {
139         //     networkAnimator.Animator.SetFloat("AnimationSpeed", moveInput);
140         // }
```

To this

```
132         // if (animator != null)
133         // {
134         //     animator.SetFloat("AnimationSpeed", moveInput);
135         // }
136
137         if (networkAnimator != null && networkAnimator.Animator != null)
138         {
139             networkAnimator.Animator.SetFloat("AnimationSpeed", moveInput);
140         }
```

8. In Hover Vehicle Controller you have to put the Network Animator like this:



9. Now you can run the game and look at the wheels of other players. They rotate!

NetworkVariable

1. Navigate to the Project window and go to Assets → Objects. Find and select HoverVehicleRoot.
2. Look in the inspector and click on the Script TankHealth.

3. Comment line 22. Keep the supplied health NetworkVariable declared.

```
22      //private int healthWithoutNetworkVariable = 100;
        0 references
23      readonly NetworkVariable<int> health = new NetworkVariable<int>(
24        100, NetworkVariableReadPermission.Everyone, NetworkVariableWritePermission.Server);
```

4. Replace all “healthWithoutNetworkVariable” with health.Value (rule: 41, 48, 102, 104, 127, 196, 232)
5. Uncommend rule 51 and 63 like this:

```
52          health.OnValueChanged += OnHealthChanged;
```

```
64          health.OnValueChanged -= OnHealthChanged;
```

6. Now you can run the game and look at the health of other players